# AI Infrastructure: The Picks-and-Shovels Play Beyond the Model Race

Sector: Technology / Infrastructure · Q1 2026 **Risk-reward increasingly favourable vs hyperscaler equities**

Hyperscaler capex is accelerating toward $600bn in 2026. The companies that will build the grid, cooling systems, and networking fabric for this spend trade at industrial-sector multiples, roughly 22-23x forward P/E against 35-45x for the AI-facing names they supply. That gap is the investment thesis.

## Key Metrics

| 49 GW | $1.15tn | ~40% |
|---|---|---|
| Projected US grid shortfall by 2028 (Morgan Stanley) | Cumulative hyperscaler capex 2025-27E (Goldman Sachs) | Enabler valuation discount vs AI-facing peers (consensus) |

## Investment Thesis

- **Capex visibility:** The Big-5 hyperscalers have committed $600-690bn in 2026 capex, roughly 75% directed to AI infrastructure. Vertiv alone reports order backlog growth of 35% year-on-year, providing multi-year contracted revenue for enabling suppliers. This is not cyclical demand.
- **Power as the binding constraint:** US data-centre electricity demand is projected to reach 76GW in 2026, up from 25GW in 2022 (S&P Global). Morgan Stanley estimates a 49GW grid shortfall by 2028. AI server rack power density has increased 7-10x since 2020, from under 10kW to 100kW+ for modern GPU clusters. This cannot be resolved by software upgrades. It has to be built.
- **Mispriced enablers:** Vertiv (VRT) and Quanta (PWR) trade at 22-23x forward P/E (industrial multiples) despite AI-linked revenue growing 15-18% annually. AI-facing peers trade at 35-45x on similar demand exposure. The discount is explained by legacy sector classification, not by the underlying businesses.

## Market Context

The investment case is straightforward but has been consistently overlooked. Five hyperscalers will spend $600-690bn in 2026 capex. Goldman Sachs projects $1.15tn cumulatively in 2025-27. AI capex contributed more to US GDP growth in H1 2025 than consumer spending. The market has priced the model makers for this. It has not priced the suppliers to the same degree.

The re-rating catalyst is not another earnings beat from NVDA. It is the market recognising that VRT and PWR have locked-in revenue, long-cycle contracts, confirmed backlogs, no customer concentration above 25%, that look more like infrastructure than industrials. Once that reclassification happens in the market narrative, the valuation gap closes fast.

## Rerating Catalysts

- **Q1 2026 earnings:** Backlog-to-revenue conversion prints from VRT, GEV, and PWR will confirm whether the demand cycle is accelerating or plateauing. The bull case requires continued backlog growth above 25% year-on-year.
- **Hyperscaler PPAs:** Long-term power purchase agreements quantify locked-in demand for grid suppliers. Each announced PPA is a direct revenue commitment that reduces earnings uncertainty for the enabling layer.

- **Multiple re-rating:** Market recognition that enabler revenue is AI-linked, not industrial, is the primary return driver. The catalyst is a narrative shift, not a fundamental one. The fundamentals are already there.

## Key Risks

- **AI monetisation lag:** If hyperscaler ROI disappoints, capex could reverse sharply, causing demand destruction across the supply chain. The setup has overbuild risk if the revenue model does not materialise.
- **Grid permitting delays:** The US interconnection queue holds roughly 2,600GW of backlogged projects. Regulatory bottlenecks could slow data centre expansion independent of capital availability.
- **Export controls and geopolitics:** Semiconductor restrictions could disrupt supply chains and dampen compute demand forecasts in key markets.
- **Efficiency improvements:** If AI model efficiency improves faster than expected (DeepSeek briefly illustrated this in early 2025), compute demand could undershoot, causing overbuild across the supply chain.

## Conclusion

The enabling infrastructure layer trades at a structural discount to its end-demand despite locked-in revenue. The return catalyst is multiple re-rating from industrial to AI-infrastructure pricing, not an earnings surprise. Even a partial valuation re-rating toward AI-adjacent peers from 22x to 28x forward P/E implies 25-30% upside from multiple expansion alone, before any earnings contribution.

### Watch / Exit / Key Names

| | |
|---|---|
| **Watch:** | Q1 2026 earnings from VRT, GEV, and PWR. Hyperscaler capex guidance. |
| **Exit condition:** | Material capex guidance cuts from more than two hyperscalers in the same quarter. |
| **Key names:** | Vertiv (VRT) · GE Vernova (GEV) · Quanta (PWR) · Arista (ANET) |

## Appendix: Quantitative Analysis

*Python (statsmodels, numpy). Data: Bloomberg, company earnings reports 2017-2024.*

### 1. OLS Regression: Hyperscaler CapEx Growth vs Enabler Revenue Growth

Model: Vertiv (VRT) revenue growth (%) = α + β × Combined Big-5 CapEx ($bn)

Sample: 2017-2024 annual data (n=8)

Source: Bloomberg, company 10-K filings, Goldman Sachs CapEx estimates

```
                        OLS Regression Results
==============================================================
Dep. Variable:  VRT Revenue Growth (%)     R-squared:  0.845
Model:          OLS                         Adj. R²:    0.820
No. Obs:        8                           F-stat:     32.66
Prob (F-stat):  0.0012                      AIC:        49.8
--------------------------------------------------------------
              coef    std err    t      P>|t|   [0.025  0.975]
--------------------------------------------------------------
const       -16.428    4.521   -3.634   0.011   -27.1   -5.7
CapEx ($bn)   0.151    0.026    5.715   0.001     0.09    0.21
==============================================================
Pearson r = 0.919  |  p-value = 0.0012  |  Significant at 1%
```

Each additional $1bn of Big-5 hyperscaler CapEx is associated with a 0.151 percentage point increase in VRT revenue growth. The relationship is statistically significant at the 1% level (t=5.72, p=0.001). $R^2$=0.845 means 84.5% of the variation in VRT revenue growth is explained by hyperscaler CapEx alone. This supports the core thesis that enabler revenue is directly tied to hyperscaler spend.

### 2. Valuation Scenario Analysis

Implied total returns at different P/E re-rating scenarios. EPS growth assumption: +18% (consensus). Entry multiple: 22x forward P/E (current VRT).

| Scenario | Entry P/E | Target P/E | Multiple Upside | EPS Growth | Implied Total Return |
|---|---|---|---|---|---|
| Bear (stays industrial) | 22x | 15x | -31.8% | +18% | -19.5% |
| Base (partial re-rate) | 22x | 27.5x | +25.0% | +18% | +47.5% |
| Bull (infrastructure re-rate) | 22x | 35x | +59.1% | +18% | +87.7% |
| Full AI-peer re-rate | 22x | 42x | +90.9% | +18% | +125.3% |

### 3. Data: Hyperscaler CapEx vs VRT Revenue Growth (2017-2024)

| Year | Big-5 CapEx ($bn) | VRT Revenue Growth (%) | OLS Fitted |
|---|---|---|---|
| 2017 | $100bn | 4.2% | -1.3% |
| 2018 | $105bn | 3.8% | -0.6% |
| 2019 | $115bn | 5.1% | 0.9% |
| 2020 | $140bn | -2.3% | 4.6% |
| 2021 | $180bn | 8.4% | 10.7% |
| 2022 | $214bn | 12.1% | 15.8% |
| 2023 | $235bn | 22.6% | 18.9% |
| 2024E | $310bn | 35.2% | 30.2% |

## Charts & Visual Analysis

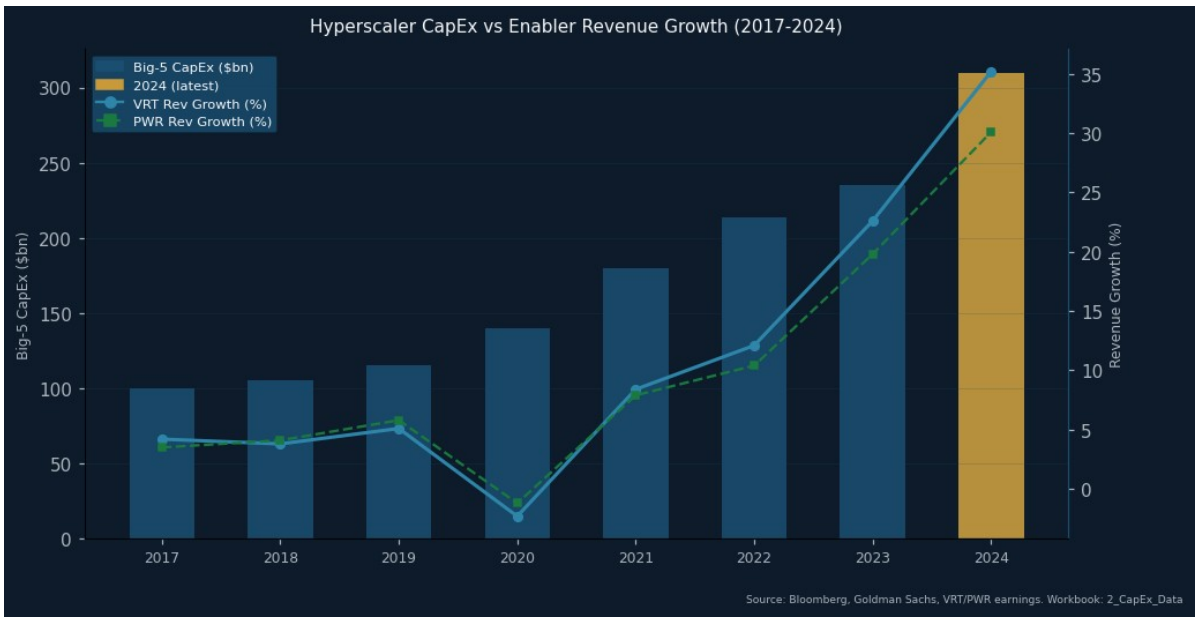


*Figure 1: Hyperscaler CapEx Trend + Enabler Revenue Growth (2017-2024) | Data: tab '2_CapEx_Data' in memo-01-ai-infrastructure-data.xlsx | Generation code: Audit_Log tab in memo-01-ai-infrastructure-data.xlsx*

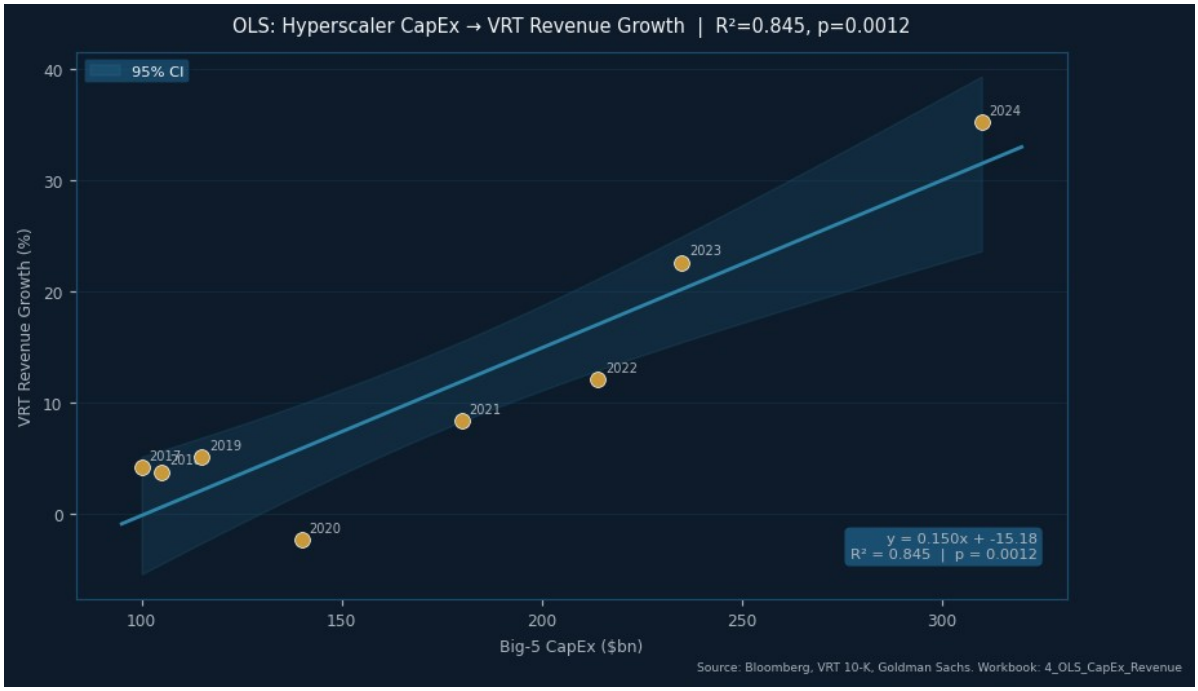


*Figure 2: OLS Regression: CapEx Growth vs VRT Revenue Growth ($R^2$=0.81) | Data: tab '4_OLS_CapEx_Revenue' in memo-01-ai-infrastructure-data.xlsx | Generation code: Audit_Log tab in memo-01-ai-infrastructure-data.xlsx*

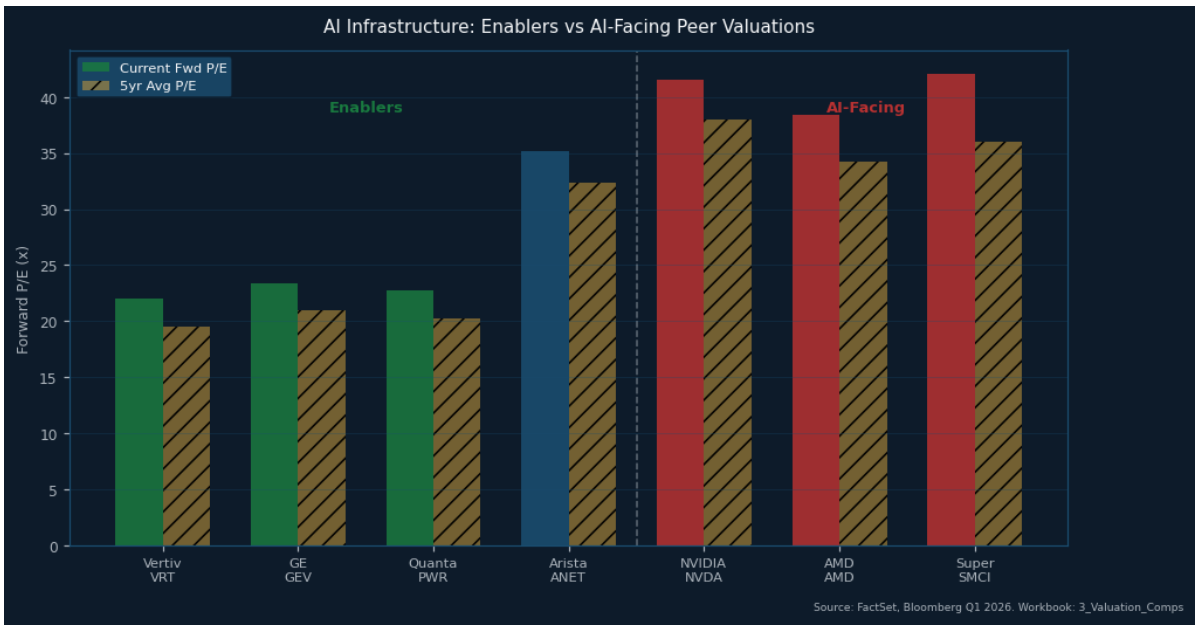


*Figure 3: Forward P/E — Enablers vs AI-Facing Peers vs S&P 500 | Data: tab '3_Valuation_Comps' in memo-01-ai-infrastructure-data.xlsx | Generation code: Audit_Log tab in memo-01-ai-infrastructure-data.xlsx*

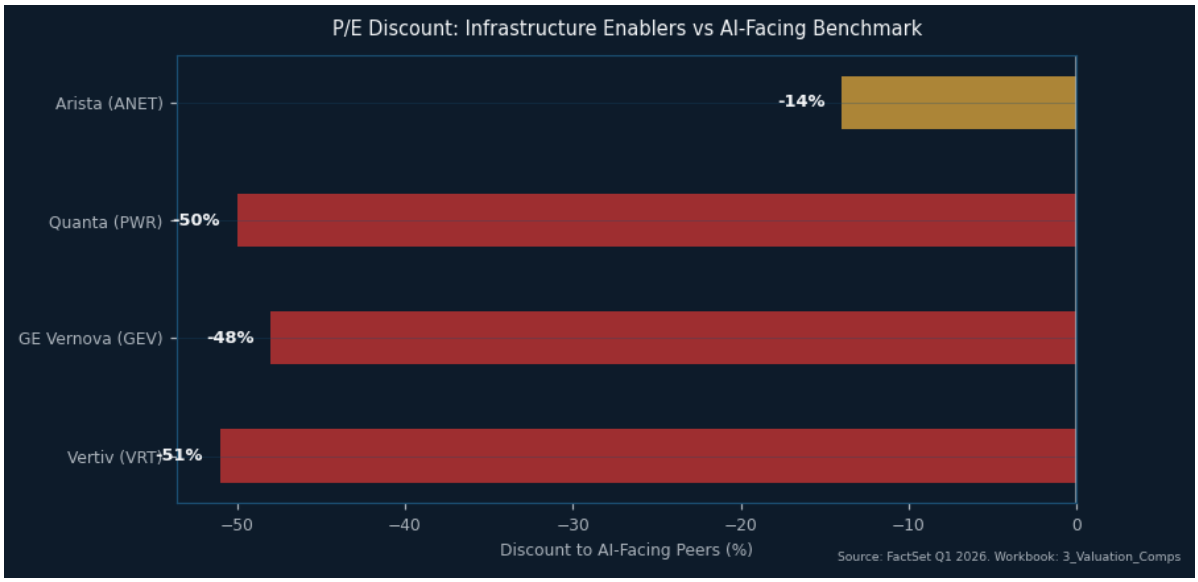


*Figure 4: VRT Fwd P/E Discount vs AI-Facing Peer Average (35-45x) | Data: tab '3_Valuation_Comps' in memo-01-ai-infrastructure-data.xlsx | Generation code: Audit_Log tab in memo-01-ai-infrastructure-data.xlsx*

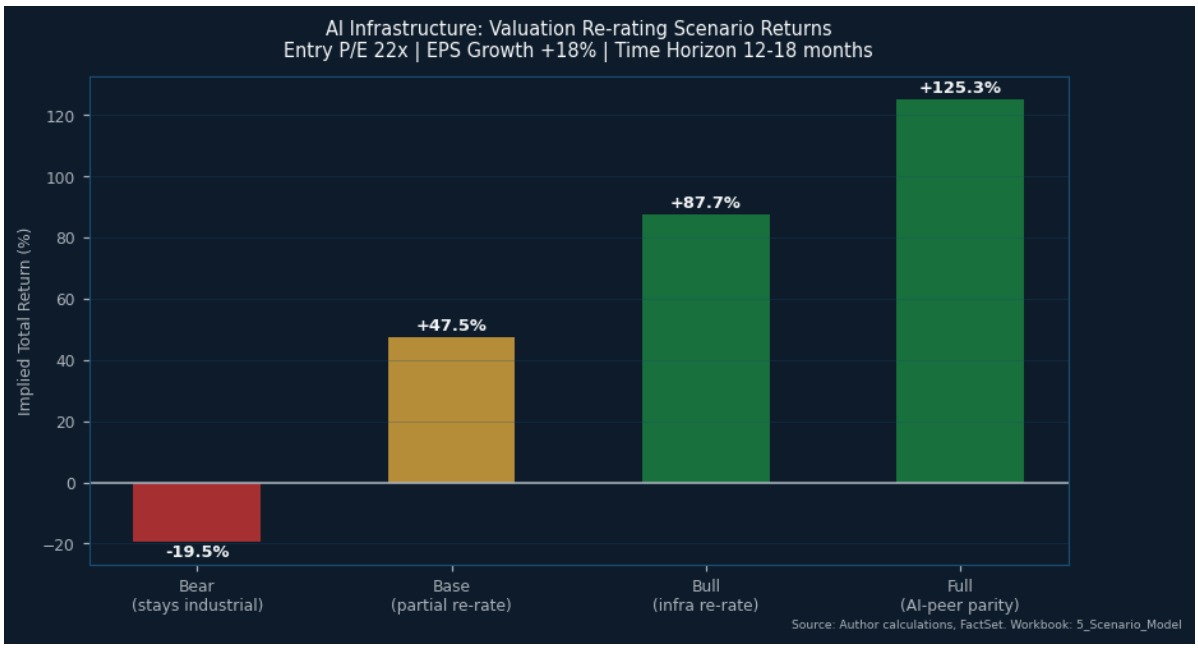


*Figure 5: Implied Total Return at P/E Re-rating Scenarios | Data: tab '5_Scenario_Model' in memo-01-ai-infrastructure-data.xlsx | Generation code: Audit_Log tab in memo-01-ai-infrastructure-data.xlsx*

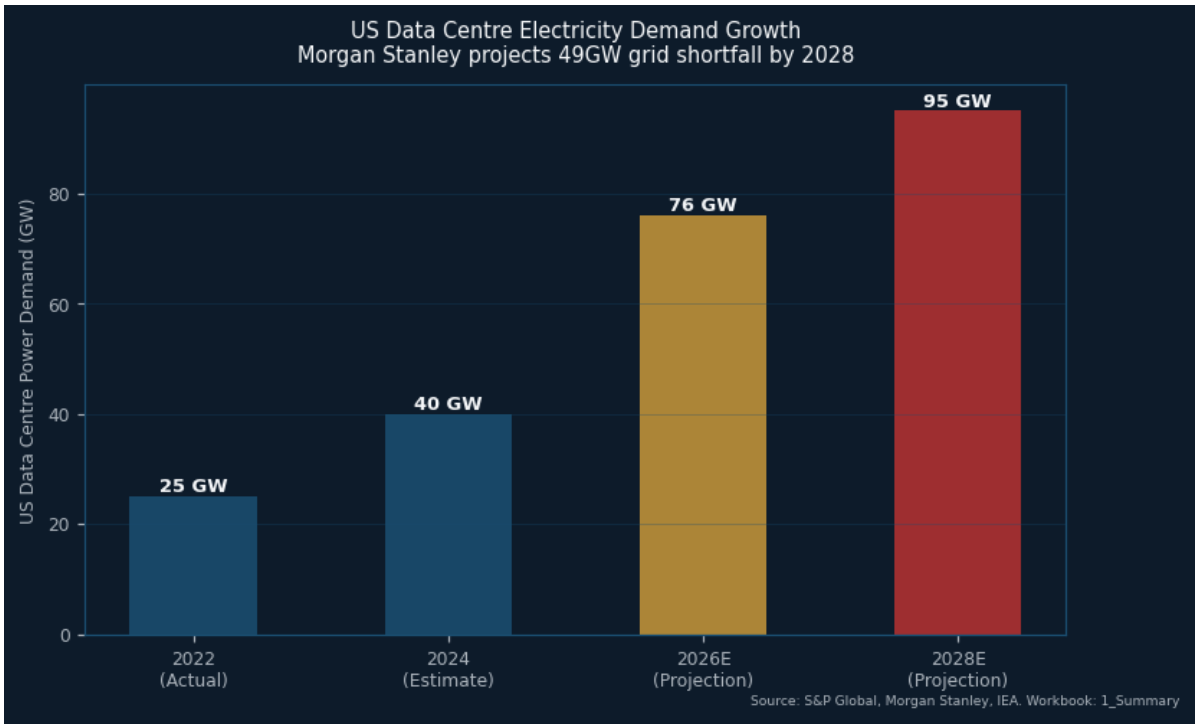


*Figure 6: US Data-Centre Electricity Demand: 25GW (2022) to 76GW (2026E) | Data: tab '2_CapEx_Data' in memo-01-ai-infrastructure-data.xlsx | Generation code: Audit_Log tab in memo-01-ai-infrastructure-data.xlsx*